

SMILE²: Estimation–Evolution via Spectral Wave Propagation for Snapshot Compressive Imaging

Anonymous submission

Abstract

Coded aperture snapshot spectral imaging (CASSI) reconstructs three-dimensional hyperspectral images from a single two-dimensional coded measurement, resulting in a severely ill-posed spatial–spectral inverse problem. While existing Transformer-based deep networks have achieved impressive performance, their core spatial–spectral interactions rely on implicit token mixing or generic transformations, lacking an explicit account of how information propagates across the hyperspectral field. Moreover, global pairwise interaction incurs quadratic computational complexity. To address these limitations, we propose SMILE² (Spectral Modulated Imaging via Learned Estimation–Evolution), a novel framework that explicitly formulates CASSI reconstruction as a two-stage process: spectral-field estimation and structured field evolution. The estimation stage constructs a dirty shift-back field, a normalized residual-adjoint cue, and shifted-mask sensing information from the forward model to produce a cleaner initial spectral field. The evolution stage then propagates this estimated field into the final hyperspectral reconstruction. At the core of both stages lies the Spectral Wave Propagator (SWP), which adapts a learnable anisotropic three-dimensional damped-wave model to the sampled spatial–spectral field and performs mask-conditioned global interaction through a closed-form Fourier-domain solution with $O(N \log N)$ operator-level complexity. Extensive experiments on both simulated and real datasets demonstrate that SMILE² achieves state-of-the-art reconstruction quality with compact computation. Ablations and cross-band analyses further validate the complementary roles of field estimation, structured evolution, and spectral propagation.

Code is to be released upon publication.

Introduction

Coded aperture snapshot spectral imaging (CASSI) aims to recover a three-dimensional hyperspectral image (HSI) from a single two-dimensional coded measurement. By using a coded aperture and a dispersive element, CASSI enables efficient snapshot spectral acquisition and avoids slow spatial or spectral scanning (Wagadarikar et al. 2008; Gehm et al. 2007). This optical efficiency, however, comes with a severely underdetermined inverse problem: spatial structures and spectral signatures are modulated, shifted, overlapped, and finally integrated into one compressed measurement.

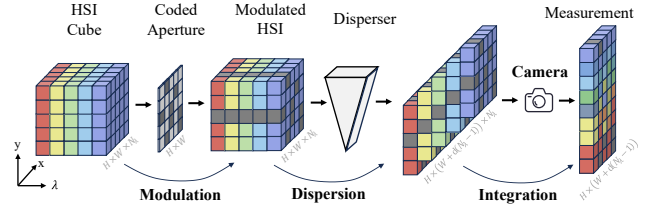


Figure 1: CASSI forward imaging. A spectral cube X is modulated by the coded aperture M_0 , dispersed into shifted spectral measurements, and compressed into a single measurement $Y = \Phi X$.

The CASSI formation process itself suggests that reconstruction should not be treated as a plain image-to-image regression problem. Because coded modulation, wavelength-dependent shift, and spectral integration are entangled in the measurement, a simple shift-back operation can expose coarse scene structures but still leaves a dirty spectral field affected by mask modulation, overlap, uneven sensing coverage, and residual inconsistency. Meanwhile, HSIs are not merely stacks of independent bands: natural spectra often exhibit strong cross-band correlation and low-dimensional material structure (Bioucas-Dias et al. 2012), while spatial edges and textures should remain coherent across wavelengths. These observations point to a field-oriented reconstruction view: estimating a cleaner initial spectral field from the multiplexed measurement, and then propagating information across the sampled spatial–spectral field.

Most end-to-end CASSI methods learn a direct mapping from the compressed measurement to the reconstructed HSI. Early neural models use convolutional or recurrent architectures to aggregate local spatial and spectral information (Miao et al. 2019; Meng, Ma, and Yuan 2020; Cheng et al. 2022), while recent Transformer-based methods further improve reconstruction quality by learning long-range spatial–spectral dependencies (Cai et al. 2022b,a,c; Luo et al. 2025). Attention is powerful because it can learn which spatial regions and spectral bands should communicate. However, this communication is usually expressed as implicit token mixing rather than an explicit propagation rule for disentangling coded spectral overlap and evolving a coherent HSI field. The scaling pressure is also visible in recent end-to-

end Transformer models: moving from MST-L to DSMT improves the average PSNR from 35.24 dB to 36.92 dB, i.e., a 1.68 dB gain, but increases the parameter count from 2.02M to 12.40M, about 6.1 times larger. Together with the quadratic core complexity of global pairwise interaction, this motivates a more structured and compact alternative for global spatial–spectral modeling.

Recent vision models suggest a complementary route: organizing feature interaction through structured dynamics instead of generic pairwise mixing. vHeat builds vision models upon heat conduction, and WaveFormer introduces wave-equation-based frequency-time visual modeling (Wang et al. 2025b; Shu et al. 2026). Frequency-domain neural operators and token mixers also show that Fourier parameterizations can support efficient global modeling (Li et al. 2021; Guibas et al. 2022; Rao et al. 2021; Lee-Thorp et al. 2022). For CASSI, a purely diffusion-like bias may oversmooth fine spatial textures and spectral variations, whereas a damped wave equation contains propagation, oscillation, and damping. In the Fourier domain, this dynamics admits a closed-form mode-wise evolution rule, making it a natural physics-inspired candidate for evolving a sampled spectral field, without claiming that the reconstructed HSI literally follows the optical propagation inside the CASSI hardware. This leads to the central question:

Can CASSI reconstruction be formulated as initial spectral-field estimation followed by evolution through a unified spatial–spectral physical operator?

This question is also aligned with a broader movement toward understanding and designing deep models through mathematical structures and dynamical systems. More broadly, recent discussions on a scientific theory of deep learning suggest that mathematical structures and dynamical descriptions can provide useful ways to understand deep models (Simon et al. 2026), while physics-guided machine learning demonstrates the value of embedding scientific structure into neural operators (Karniadakis et al. 2021; Willard et al. 2023; E 2017). This work explores such a direction for CASSI reconstruction.

We answer this question with SMILE², a single-pass estimation–evolution framework. At its core, the Spectral Wave Propagator (SWP) adapts a learnable three-dimensional damped-wave model to the sampled HSI field and performs structured global spatial–spectral interaction through a Fourier-domain closed-form solution. CASSI preparation then derives a dirty shift-back field, a measurement-coverage normalization map, a residual back-projection, and shifted-mask information from the forward imaging model. Based on these measurement-consistent quantities, SFESTIMATOR estimates a cleaner initial spectral field, and SFEVOLVER further propagates it into the final HSI reconstruction. Compared with the $O(N^2)$ pairwise token mixing of self-attention, SWP provides global interaction through an $O(N \log N)$ operator-level Fourier-domain computation.

Our contributions are summarized as follows.

- We formulate end-to-end CASSI reconstruction as spectral-field estimation followed by physical evolution,

providing a single-pass alternative to implicit token mixing.

- We design SWP, a mask-conditioned damped-wave propagation operator that performs structured global spatial–spectral interaction through a Fourier-domain closed-form wave kernel.
- We connect the CASSI forward model with spectral-field propagation through CASSI preparation, SFESTIMATOR, and SFEVOLVER, enabling dirty-field correction before structured evolution.
- We build compact SMILE-S/M/L variants and obtain state-of-the-art end-to-end reconstruction quality on the TSA simulation benchmark, with favorable SAM, visual fidelity, and mechanism analyses.

Related Work

End-to-End CASSI Reconstruction

Early CASSI reconstruction relies on model-based optimization with hand-crafted priors, which provides an interpretable inverse formulation but requires iterative inference and careful parameter tuning (Bioucas-Dias and Figueiredo 2007; Yuan 2016; Liu et al. 2019). Learning-based reconstruction subsequently develops along direct end-to-end and model-driven unfolding lines. End-to-end methods learn a single-pass mapping from the coded measurement to the HSI, enabling substantially faster inference. Representative designs include convolutional regression in λ -Net, spatial–spectral attention in TSA-Net, bidirectional recurrence in BIRNAT, and high-resolution dual-domain learning in HD-Net (Miao et al. 2019; Meng, Ma, and Yuan 2020; Cheng et al. 2022; Hu et al. 2022).

Transformer-based end-to-end methods further improve reconstruction by modeling long-range spectral correlations. MST performs mask-guided spectral-wise attention, whereas CST and MST++ improve spectral interaction through sparse and multi-stage designs (Cai et al. 2022b,a,c). SPECAT (Yao et al. 2024) progressively accumulates spatial and spectral attention, while S2-Transformer (Wang et al. 2025a) strengthens mask-aware spatial–spectral interaction. DSMT (Luo et al. 2025) adopts coarse-to-fine reconstruction with dual-window multiscale attention, and LSST (Li, Zhao, and Xu 2026) separates lightweight spectral and spatial processing. Despite these advances, spatial–spectral communication is still implemented through learned convolutional or attention-based mixing, without an explicit rule describing how information propagates over the sampled HSI field. SMILE² addresses this gap by coupling the x , y , and λ axes through mask-gated damped-wave evolution in the Fourier domain.

Forward-Model Cues in End-to-End Reconstruction

Although end-to-end networks learn a direct mapping from measurement to HSI, the CASSI forward model remains essential for network design. GAP-Net embeds projection and back-projection into the network, and DGSMP constructs deep priors from the sensing matrix structure (Meng, Jalali,

and Yuan 2020; Huang et al. 2021). These works show that the sensing operator, shifted mask, and measurement-adjoint cues provide useful initialization and correction information. Multi-stage unfolding methods further exploit the forward model through repeated data-consistency and learned-prior updates, often achieving higher absolute PSNR at higher iterative cost (Cai et al. 2022d; Zhang et al. 2024a,b). SMILE² builds on the broader insight that forward-model cues matter in a single-pass end-to-end framework. CASSI preparation condenses measurement-aware cues into H_0 , P , and Q_0 , enabling SFESTIMATOR and SFEVOLVER to perform field correction and propagation within one feed-forward reconstruction process.

Frequency- and PDE-Inspired Vision Modeling

Frequency-domain modeling offers a different route to global interaction. Fourier Neural Operator learns solution operators in function spaces and demonstrates that Fourier modes can efficiently parameterize global mappings (Li et al. 2021). FNet mixes tokens with Fourier transforms, GFNet learns global filters, and AFNO adapts neural operator ideas to high-resolution vision features (Lee-Thorp et al. 2022; Rao et al. 2021; Guibas et al. 2022). In image restoration, FFTformer and frequency-modulated restoration networks use Fourier-domain attention or frequency modulation to model degradation-related components (Kong et al. 2023; Cui et al. 2024). These studies indicate that frequency transforms can provide global receptive fields without explicitly constructing quadratic token affinities.

Another related direction is PDE-inspired vision modeling. vHeat builds vision models upon heat conduction, while WaveFormer uses a wave equation to design frequency-time decoupled visual modeling (Wang et al. 2025b; Shu et al. 2026). Compared with total variation, which mainly constrains local first-order gradients, and compared with heat diffusion, which is biased toward smoothing, a damped wave equation contains propagation, oscillation, and damping. This makes it suitable for modeling second-order spatial-spectral structures and frequency-selective attenuation. In spectral reconstruction, Phy-CoSF introduces physics-guided continuous spectral fields for snapshot compressive imaging reconstruction and super-resolution (Chen et al. 2026). However, these frequency- and PDE-inspired works do not directly address CASSI-specific mask modulation, wavelength-dependent shift, 2D-to-3D compression, and shift-back initialization. SMILE² differs by coupling CASSI preparation with a mask-gated damped-wave spectral-field propagator in a fixed-grid, single-pass CASSI reconstruction framework.

Method

Spectral-Field Formulation and Notation

Let $X \in \mathbb{R}^{H \times W \times L}$ denote the target HSI sampled on a fixed spatial-spectral grid, where the axes correspond to x , y , and wavelength λ . Let $Y \in \mathbb{R}^{H \times W'}$ denote the 2D CASSI measurement, where $W' = W + d(L - 1)$ when adjacent spectral channels are shifted by d pixels. We denote the physical

coded aperture by $M_0 \in \mathbb{R}^{H \times W}$, and its replicated spectral stack by $M \in \mathbb{R}^{H \times W \times L}$. The complete sensing operator is denoted by Φ , which includes mask modulation, wavelength-dependent spatial shift, and spectral summation. The shifted mask stack in the measurement plane is denoted by $\Phi_s^Y \in \mathbb{R}^{H \times W' \times L}$, while its field-width crop used by SWP is denoted by $G_s \in \mathbb{R}^{H \times W \times L}$. The shift-back field, residual-adjoint cue, estimated spectral field, and final reconstruction are denoted by H_0 , Q_0 , U_1 , and $U_2 = \hat{X}$, respectively. We use ϵ as a small positive constant for numerical stability and gate lower bounds.

CASSI Forward Model and Field Preparation

The CASSI measurement is obtained by band-wise coding, wavelength-dependent shifting, and spectral summation:

$$Y = \Phi X = \sum_{\lambda=1}^L \text{Shift}_{\lambda}(M_{\lambda} \odot X_{\lambda}). \quad (1)$$

The normalization map is

$$P = \Phi \Phi^{\top} = \sum_{\lambda=1}^L (\Phi_{s,\lambda}^Y)^2, \quad (2)$$

In vectorized notation, Φ maps a 3D spectral field to the 2D measurement plane; hence $\Phi \Phi^{\top}$ corresponds to a diagonal normalization operator in the measurement domain. In implementation, the full matrix is not stored. Instead, $P \in \mathbb{R}^{H \times W'}$ is represented by its diagonal map, which summarizes the shifted mask energy contributing to each measurement pixel. For a binary mask, P can be interpreted as the effective coding coverage at each measurement location. It is therefore not a global scalar but a pixel-wise measurement-domain weight map. The shifted-mask cue used by SWP is obtained from the measurement-domain shifted mask stack and cropped to the field width:

$$\begin{aligned} \Phi_s^Y &= \text{Shift}(M) \in \mathbb{R}^{H \times W' \times L}, \\ G_s &= \text{Crop}_W(\Phi_s^Y) \in \mathbb{R}^{H \times W \times L}. \end{aligned} \quad (3)$$

We first construct a dirty shift-back field:

$$\begin{aligned} H_{0,\lambda} &= \text{Back}_{\lambda}(Y), \\ H_0 &= \text{Back}(Y) \in \mathbb{R}^{H \times W \times L}, \end{aligned} \quad (4)$$

where Back_{λ} crops and shifts a measurement-domain image back to the spatial coordinates of the λ -th spectral channel, and Back stacks all L channel-wise results. H_0 contains major scene structures but remains an aliased spectral field affected by mask modulation, dispersion overlap, and spectral ambiguity. We further compute a residual-adjoint cue:

$$R_0 = Y - \Phi H_0, \quad Q_0 = \Phi^{\top}(R_0 / (P + \epsilon)). \quad (5)$$

The input to SFESTIMATOR is

$$C_0 = [H_0, Q_0, M, H_0 \odot M] \in \mathbb{R}^{H \times W \times 4L}. \quad (6)$$

Here Q_0 exposes the measurement residual left by the initial shift-back field, while $H_0 \odot M$ explicitly couples image content with the coded aperture.

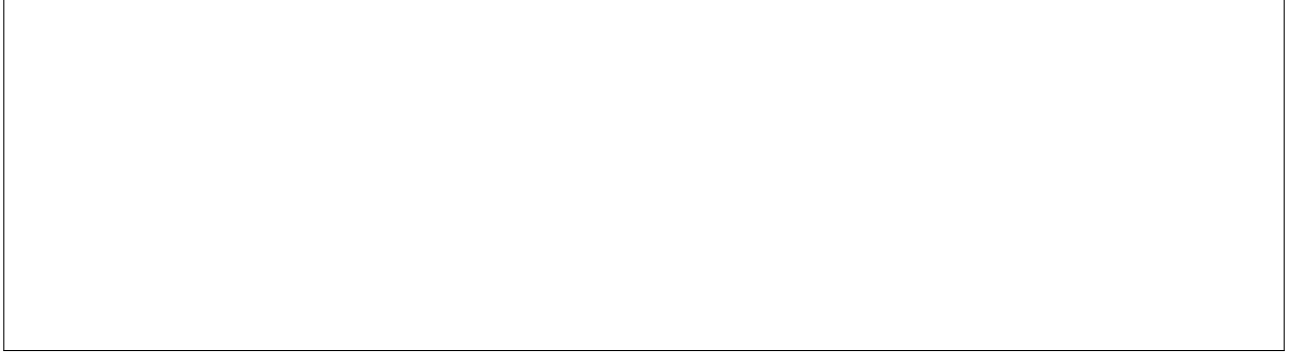


Figure 2: Conceptual overview of the transition from attention mixing to physical evolution. SMILE² replaces generic pairwise token interaction with structured spatial–spectral propagation governed by a Fourier-domain damped-wave kernel.

Spectral Wave Propagator

SWP is the core physical evolution operator used in both SFESTIMATOR and SFEVOLVER. Given a spectral field $z \in \mathbb{R}^{H \times W \times L}$, SWP regards it as a field over (x, y, λ) and performs learnable damped-wave propagation. The evolution time t denotes feature-field propagation depth, rather than exposure time or physical travel time in the CASSI optical path. Thus, SWP is a physics-inspired regularized propagation operator for reconstruction, not a direct simulation of the optical hardware.

To condition propagation on the sensing process, SWP first gates the input field with the field-aligned shifted mask cue:

$$\tilde{z} = (\epsilon + (1 - \epsilon)G_s) \odot z. \quad (7)$$

Two lightweight learnable mappings then generate the initial displacement and velocity of the wave equation:

$$u_0 = \Pi_u(\tilde{z}), \quad v_0 = \Pi_v(\tilde{z}). \quad (8)$$

Applying a real-valued 3D Fourier transform over x, y, λ gives

$$\hat{u}_0 = \mathcal{F}_{x,y,\lambda}(u_0), \quad \hat{v}_0 = \mathcal{F}_{x,y,\lambda}(v_0). \quad (9)$$

The Fourier transform along the spectral axis is used as a discrete spectral-mode basis over the sampled HSI bands, not as an assumption that real spectral reflectance is physically periodic.

The physical backbone of SWP is the anisotropic 3D damped wave equation:

$$\partial_{tt}u + \alpha\partial_tu = v_s^2(\partial_{xx} + \partial_{yy})u + v_\lambda^2\partial_{\lambda\lambda}u. \quad (10)$$

For a Fourier mode $\mathbf{f} = (f_x, f_y, f_\lambda)$, the second-order derivatives become multiplicative factors. The PDE is converted into a per-mode second-order ordinary differential equation:

$$\frac{d^2\hat{u}(\mathbf{f}, t)}{dt^2} + \alpha(f_\lambda)\frac{d\hat{u}(\mathbf{f}, t)}{dt} + \omega_0^2(\mathbf{f})\hat{u}(\mathbf{f}, t) = 0, \quad (11)$$

where

$$\begin{aligned} \omega_0^2(\mathbf{f}) &= (2\pi)^2 [v_s^2(f_\lambda)(f_x^2 + f_y^2) + v_\lambda^2 f_\lambda^2], \\ \eta(\mathbf{f}) &= \omega_0^2(\mathbf{f}) - \left(\frac{\alpha(f_\lambda)}{2}\right)^2. \end{aligned} \quad (12)$$

The closed-form evolution is

$$\begin{aligned} \hat{u}(\mathbf{f}, t) &= e^{-\alpha(f_\lambda)t/2} \left[\hat{u}_0(\mathbf{f}) \text{Cs}(\eta(\mathbf{f}), t) \right. \\ &\quad \left. + \left(\hat{v}_0(\mathbf{f}) + \frac{\alpha(f_\lambda)}{2} \hat{u}_0(\mathbf{f}) \right) \text{Sn}(\eta(\mathbf{f}), t) \right]. \end{aligned} \quad (13)$$

Cs and Sn denote cosine-like and sine-like terms that unify under-damped and over-damped cases. For $\eta > 0$, $\text{Cs}(\eta, t) = \cos(\sqrt{\eta}t)$ and $\text{Sn}(\eta, t) = \sin(\sqrt{\eta}t)/\sqrt{\eta}$; for $\eta < 0$, they naturally become hyperbolic functions. The evolved field is obtained by inverse FFT:

$$u(t) = \mathcal{F}_{x,y,\lambda}^{-1}(\hat{u}(\mathbf{f}, t)). \quad (14)$$

The parameters $\alpha(f_\lambda)$ and $v_s(f_\lambda)$ are spectral-mode-dependent damping and propagation speed, while v_λ controls the strength of the spectral second derivative. In the implemented SWP layers, α and v_s are layer-specific learnable vectors indexed by the spectral-mode dimension, initialized as constant damping and speed values and mapped through a softplus function to keep them positive. The scalar v_λ and the evolution depth t are also learned with the same positivity constraint. Consequently, $\alpha(f_\lambda)$ and $v_s(f_\lambda)$ are treated as learnable parameters over the discrete spectral-mode basis. The output of SWP is written as a residual update:

$$\mathcal{W}_\theta(z, G_s) = z + \Pi_o(u(t)). \quad (15)$$

Self-attention explicitly computes pairwise token relations with $O(N^2)$ core complexity. SWP instead performs global propagation through FFT and a closed-form wave kernel, yielding $O(N \log N)$ operator-level complexity.

Spectral Field Estimator

The aliased shift-back field H_0 is not directly suitable as the final reconstruction. SFESTIMATOR first predicts a zero-initialized residual correction:

$$\Delta_\psi(C_0) = \text{Conv}_{3 \times 3}^0(\sigma(\text{Conv}_{3 \times 3}(C_0))), \quad (16)$$

where $\text{Conv}_{3 \times 3}^0$ denotes a zero-initialized 3×3 convolution. Therefore, at the beginning of training,

$$Z_0 = H_0 + \Delta_\psi(C_0) = H_0, \quad (17)$$

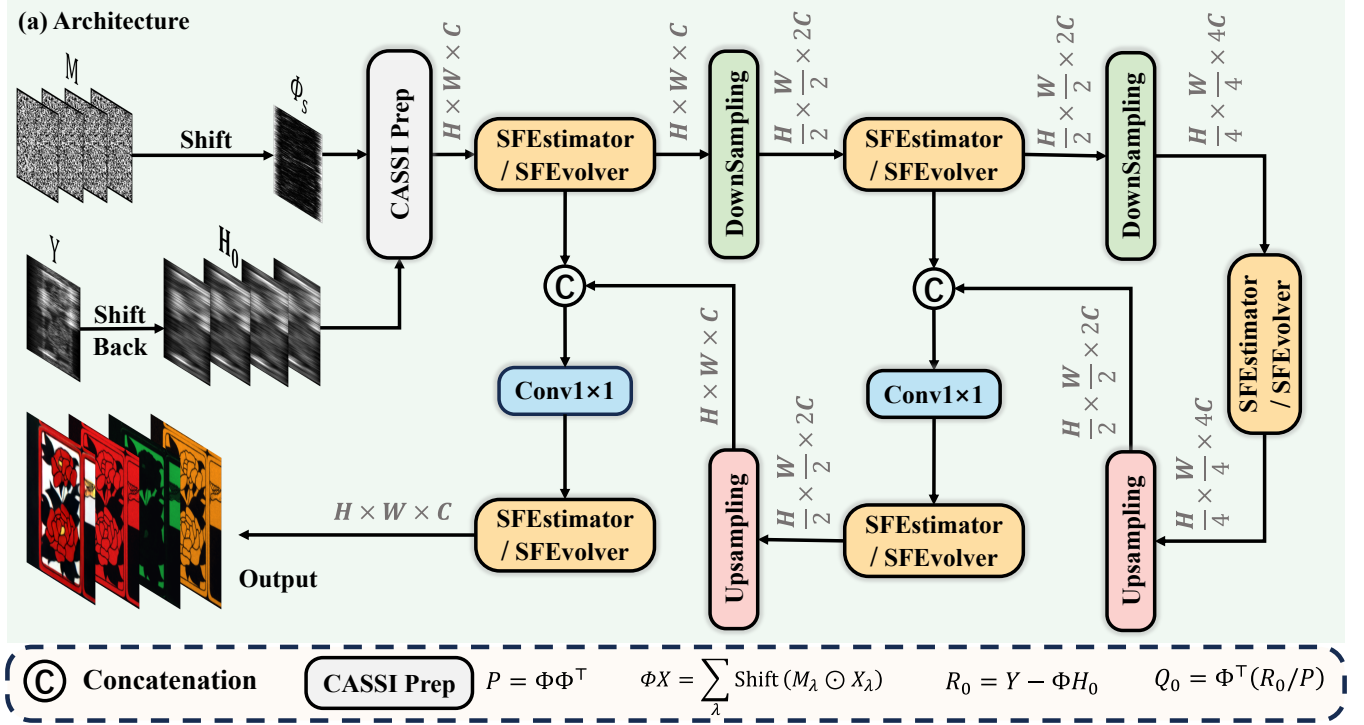


Figure 3: Overall architecture of SMILE². CASSI preparation constructs H_0 , Q_0 , and the field-aligned shifted-mask cue G_s from the measurement and mask. SFESTIMATOR estimates the initial spectral field U_1 , and SFEVOLVER evolves it into the final reconstruction $U_2 = \hat{X}$.

so the network does not destroy the shift-back initialization. The first SWP-based U-Net estimates a cleaner spectral field:

$$U_1 = \mathcal{E}_{\theta_E}^{\text{SWP}}(Z_0, G_s). \quad (18)$$

In the architecture, the SFE stem and the first SWP-based U-Net together constitute SFESTIMATOR.

Spectral Field Evolver

SFEVOLVER does not recompute R_0 or Q_0 . Instead, it further evolves the estimated spectral field U_1 :

$$U_2 = \mathcal{E}_{\theta_V}^{\text{SWP}}(U_1, G_s), \quad \hat{X} = U_2. \quad (19)$$

The main signal flow is

$$Y, M_0 \rightarrow H_0, Q_0, M, G_s \rightarrow U_1 \rightarrow \hat{X}. \quad (20)$$

We instantiate three compact variants to evaluate the same estimation–evolution design under different computational budgets: SMILE-S uses the compact $[2, 2, 2]$ backbone with partially tied estimator/evolver parameters, SMILE-M decouples the estimator and evolver parameters at the same width, and SMILE-L increases the U-Net width to $[2, 4, 4]$ for higher capacity.

Experiments

Experimental Settings

Simulation experiments are conducted on the TSA simulation benchmark with ten test scenes of spatial size 256×256

and $L = 28$ spectral channels. Training data are sampled from CAVE and KAIST hyperspectral datasets (Yasuma et al. 2010; Choi et al. 2017). We report PSNR, SSIM, and spectral angle mapper (SAM), where SAM measures spectral directional error and is important for hyperspectral fidelity (Kruse et al. 1993). All training and evaluation experiments are conducted on NVIDIA A800 GPUs with 80GB memory. The main models are trained for 300 epochs with Adam and cosine learning-rate decay.

SFESTIMATOR and SFEVOLVER output U_1 and $U_2 = \hat{X}$, respectively. Training uses only weighted RMSE reconstruction supervision:

$$\mathcal{L}_{\text{train}} = \sum_{k=1}^2 \beta_k \text{RMSE}(U_k, X), \quad \beta = [0.2, 1.0]. \quad (21)$$

The objective does not include auxiliary SAM/SSIM, spectral-derivative, spectral-angle, or explicit $\|\Phi \hat{X} - Y\|$ data-fidelity losses. Forward-model consistency enters before reconstruction through CASSI preparation: H_0 supplies the dirty shift-back field, Q_0 exposes the normalized residual-adjoint cue, and P represents the measurement-domain coding coverage.

Quantitative Comparison

Table 1 reports per-scene results on the TSA simulation benchmark. For methods with released weights or reproducible official implementations, we evaluate them under

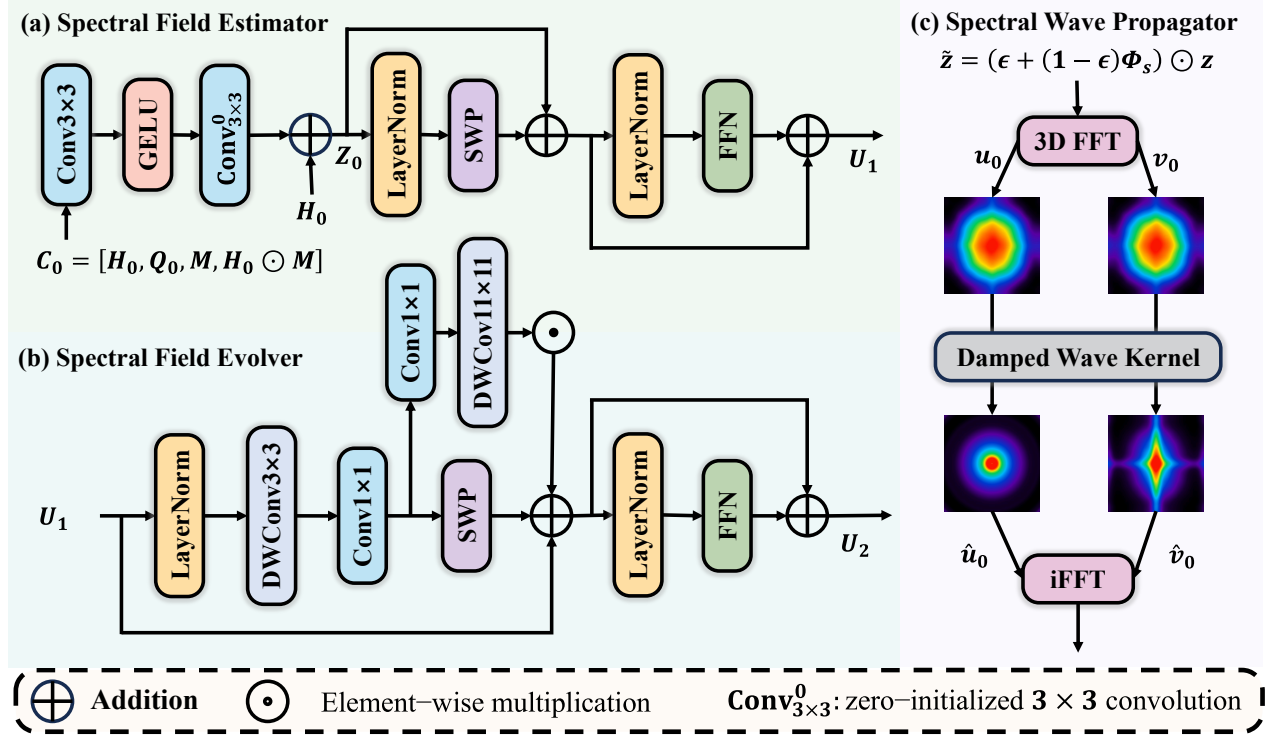


Figure 4: Signal flow of SFESTIMATOR, SFEVOLVER, and the embedded SWP operator. The SWP branch performs mask-gated Fourier-domain damped-wave propagation, while SFESTIMATOR and SFEVOLVER differ in their input fields and update roles.

the same protocol; for methods without released weights or reconstructed outputs, PSNR/SSIM are taken from the corresponding papers and SAM is denoted by “–”. Compared with earlier CNN and Transformer E2E baselines, SMILE² consistently improves reconstruction quality while keeping the model compact. SMILE²-M uses 2.00M parameters and 28.28G FLOPs, reaching 36.57 dB PSNR, 0.961 SSIM, and 0.097 rad SAM. Scaling the same estimation–evolution principle to SMILE²-L improves the average score to 37.07 dB PSNR, 0.965 SSIM, and 0.086 rad SAM with 3.19M parameters and 38.60G FLOPs. These results indicate that structured spectral-field evolution can serve as an effective E2E reconstruction prior, improving both image-level fidelity and spectral-angle consistency.

Qualitative Results

Figures 5 and 6 show representative simulation results. Scene 5 contains weak text boundaries in the “SIG” region, where low-accuracy methods recover the rough layout but introduce artifacts and spectral distortions. Transformer baselines are cleaner, yet local structures are still softened. SMILE²-M and SMILE²-L preserve sharper letters and local textures, and their spectral density curves more closely follow the ground truth. Scene 7 contains strong spatial–spectral contrast. SMILE² maintains clearer petal boundaries across multiple bands and suppresses dispersion-like artifacts. Together with the SAM scores in Table 1, these visual results indicate that the improvement is reflected not

only in average image quality but also in spectral consistency and local structure.

We also apply SMILE² to real CASSI measurements. As shown in Fig. 7, SMILE² produces stable reconstructions on real coded measurements. Compared with other end-to-end baselines, it shows fewer artifacts in smooth regions and clearer local edges, suggesting that the estimation–evolution structure generalizes to real measurements.

Ablation Study

The ablation results show the contribution of the estimation–evolution decomposition. Removing SFESTIMATOR keeps the evolution structure but removes the initial-field correction from Q_0 and $H_0 \odot M$, causing degradation in PSNR, SSIM, and SAM. Removing SFEVOLVER forces the network to rely only on the estimated field and leads to severe quality loss, showing that structured evolution is not a simple post-processing branch. Removing shifted-mask gating or the λ -derivative term also worsens SAM, indicating that mask-conditioned propagation and spectral propagation both contribute to spectral fidelity.

Mechanism Analysis

We further analyze whether SMILE² actually exploits cross-band information. We estimate an implicit cross-band mixing map through finite-difference impulse response and compare performance drops under equal-count occlusion of neighboring and distant bands. As shown in Fig. 8,

Table 1: Quantitative comparison on the TSA simulation benchmark. Each scene cell reports PSNR, SSIM, and SAM(rad) from top to bottom. Methods with public checkpoints or official reproducible implementations are evaluated under the same protocol; for methods without released weights or reconstructed outputs, PSNR/SSIM are taken from the corresponding papers and SAM is unavailable.

Algorithm	Method	Params	GFLOPs	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Avg
Optimization	TwIST (Bioucas-Dias and Figueiredo 2007)	–	–	25.16	23.02	21.40	30.19	21.41	20.95	22.20	21.82	22.42	22.67	23.12
				0.700	0.604	0.711	0.851	0.635	0.644	0.643	0.650	0.690	0.569	0.669
				–	–	–	–	–	–	–	–	–	–	–
Optimization	GAP-TV (Yuan 2016)	4.27	75.63	21.26	21.90	18.97	27.97	17.43	21.88	20.65	22.19	22.27	19.94	21.44
				0.490	0.578	0.506	0.794	0.429	0.688	0.442	0.700	0.609	0.615	0.585
				0.553	0.618	0.561	0.455	0.504	0.548	0.530	0.580	0.366	0.594	0.531
Optimization	DeSCI (Liu et al. 2019)	–	–	27.13	23.04	26.62	34.96	23.94	22.38	24.45	22.03	24.56	23.59	25.27
				0.748	0.620	0.818	0.897	0.706	0.683	0.743	0.673	0.732	0.587	0.721
				–	–	–	–	–	–	–	–	–	–	–
CNN	λ -Net (Miao et al. 2019)	32.73	28.40	29.71	27.70	29.55	37.58	26.61	27.28	26.62	26.23	28.56	26.16	28.60
				0.832	0.746	0.852	0.919	0.804	0.809	0.783	0.807	0.805	0.723	0.808
				0.244	0.295	0.260	0.392	0.260	0.403	0.244	0.416	0.265	0.391	0.317
CNN	TSA-Net (Meng, Ma, and Yuan 2020)	44.25	73.83	24.42	23.38	17.30	26.34	20.51	22.29	21.35	22.25	20.27	21.29	21.94
				0.714	0.660	0.614	0.795	0.600	0.742	0.636	0.762	0.672	0.694	0.689
				0.256	0.315	0.445	0.415	0.275	0.368	0.374	0.395	0.446	0.442	0.373
MAP+CNN	DGSMP (Huang et al. 2021)	3.76	646.36	33.35	31.66	32.92	40.39	29.46	32.74	31.14	31.33	31.53	31.51	32.60
				0.920	0.892	0.925	0.970	0.894	0.938	0.898	0.932	0.925	0.934	0.923
				0.161	0.208	0.135	0.135	0.172	0.144	0.141	0.190	0.145	0.129	0.156
CNN	HDNet (Hu et al. 2022)	2.37	154.73	35.11	35.65	36.05	42.48	32.67	34.47	33.64	32.45	34.85	32.34	34.97
				0.940	0.943	0.948	0.978	0.950	0.956	0.930	0.948	0.947	0.943	0.948
				0.129	0.142	0.099	0.101	0.089	0.120	0.114	0.141	0.110	0.119	0.116
Transformer	MST-L (Cai et al. 2022b)	2.02	25.67	35.44	36.12	36.39	42.05	32.94	34.71	34.08	32.90	35.04	32.74	35.24
				0.946	0.949	0.955	0.977	0.950	0.957	0.932	0.953	0.947	0.946	0.951
				0.123	0.142	0.106	0.128	0.101	0.137	0.110	0.179	0.128	0.146	0.130
Transformer	CST-L (Cai et al. 2022a)	3.00	20.34	35.82	36.52	37.45	42.46	33.34	35.48	34.40	33.80	35.88	33.35	35.85
				0.951	0.955	0.963	0.979	0.956	0.966	0.940	0.965	0.957	0.955	0.959
				0.116	0.118	0.082	0.100	0.078	0.104	0.098	0.119	0.100	0.103	0.102
Transformer	S2Transformer (Wang et al. 2025a)	1.80	27.21	36.17	37.57	37.29	42.96	34.40	36.44	35.41	34.50	36.54	33.57	36.48
				0.949	0.958	0.957	0.975	0.960	0.965	0.946	0.962	0.959	0.952	0.958
				–	–	–	–	–	–	–	–	–	–	–
Transformer	DSMT (Luo et al. 2025)	12.40	44.94	36.38	37.86	39.18	43.89	34.03	36.48	35.48	34.54	37.67	33.71	36.92
				0.958	0.965	0.968	0.985	0.965	0.972	0.952	0.969	0.966	0.959	0.966
				–	–	–	–	–	–	–	–	–	–	–
Wave equation	SMILE ² -S	1.02	28.28	35.68	36.39	37.39	44.07	33.42	35.20	34.29	33.27	37.21	32.45	35.94
				0.946	0.950	0.961	0.984	0.955	0.961	0.936	0.957	0.960	0.947	0.956
				0.114	0.131	0.084	0.104	0.086	0.123	0.101	0.148	0.102	0.124	0.112
Wave equation	SMILE ² -M	2.00	28.28	36.16	37.43	38.77	43.67	33.86	35.66	35.13	33.98	37.93	33.09	36.57
				0.952	0.959	0.967	0.982	0.960	0.965	0.947	0.964	0.965	0.953	0.961
				0.106	0.113	0.070	0.089	0.074	0.106	0.090	0.128	0.089	0.104	0.097
Wave equation	SMILE ² -L	3.19	38.60	36.60	38.04	39.25	44.78	34.23	35.94	35.61	33.96	38.65	33.59	37.07
				0.957	0.966	0.969	0.987	0.964	0.969	0.952	0.966	0.969	0.956	0.965
				0.091	0.097	0.071	0.080	0.072	0.089	0.078	0.103	0.082	0.097	0.086

the model exhibits strong diagonal responses, meaningful nearby-band coupling, and weak but non-negligible long-range spectral interactions. This indicates that SMILE² induces measurable cross-band propagation with both local spectral coupling and long-range spectral context.

Discussion

Interpretability of physical evolution. Unlike direct pairwise token interaction, SWP writes spatial–spectral mixing as Fourier-domain damped-wave propagation. This gives ex-

plicit structural meaning to global propagation, the spectral second-derivative term, and CASSI mask gating. The mechanism analysis further shows that the model uses both neighboring and distant spectral contexts, which is consistent with the coexistence of spectral continuity and material-dependent variation in HSIs.

Estimation before evolution. The CASSI shift-back field contains major scene structures but also contains mask modulation, dispersion overlap, and spectral aliasing. Directly evolving H_0 is therefore suboptimal. SFESTIMA-

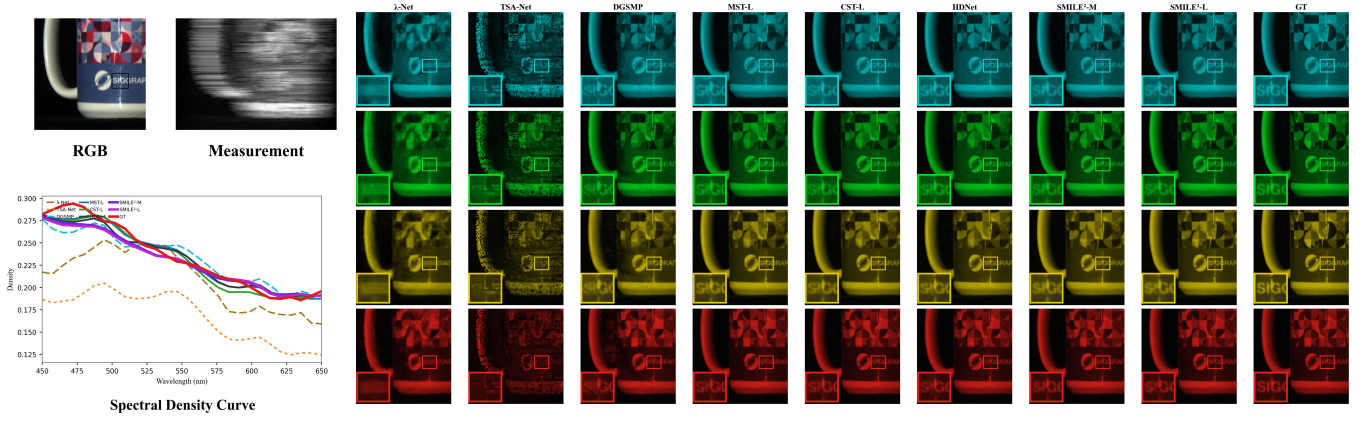


Figure 5: Simulation Scene 5. The figure compares RGB/measurement inputs, representative spectral bands, zoomed local regions, and spectral density curves.

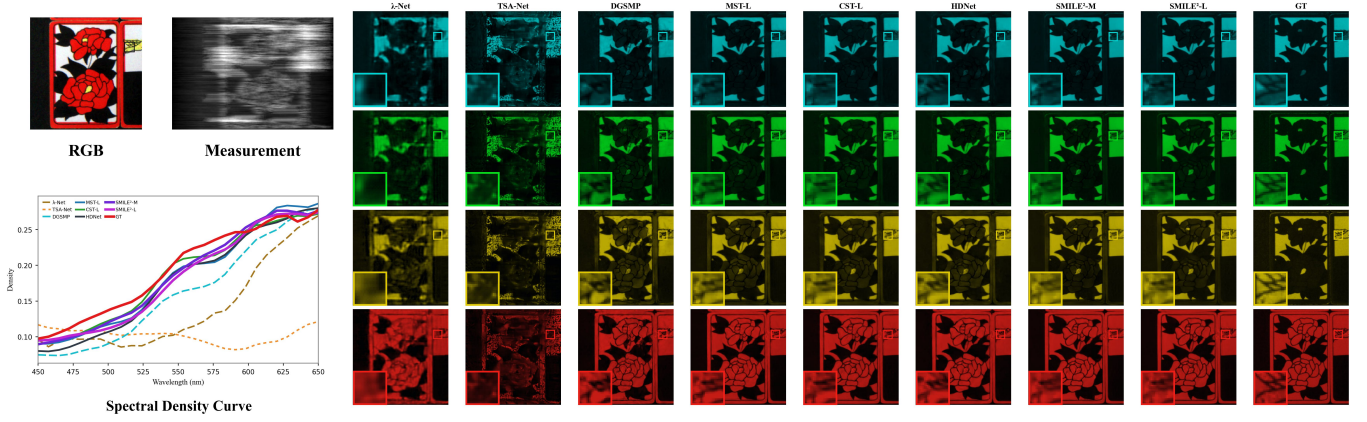


Figure 6: Simulation Scene 7. The same set of end-to-end comparison methods and curve styles are used as in Scene 5.

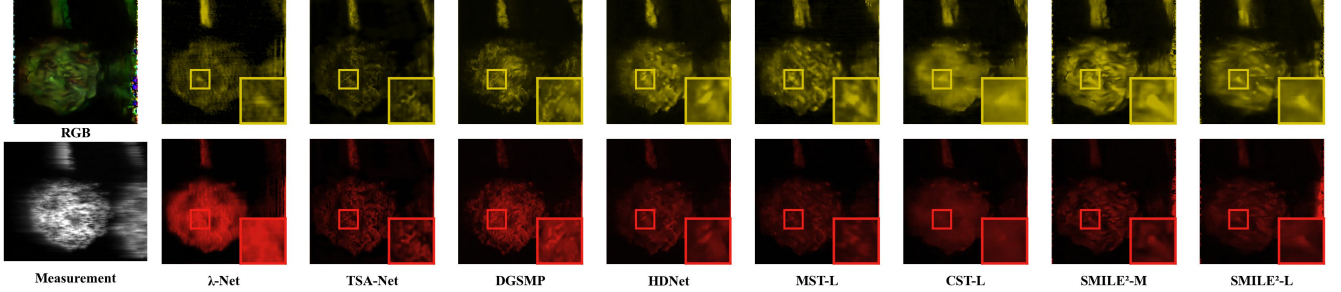


Figure 7: Real CASSI measurement results. Since real measurements do not provide ground-truth HSIs, we compare representative bands and zoomed local regions.

TOR uses Q_0 and $H_0 \odot M$ to estimate a more reliable initial spectral field, allowing SFEVOLVER to focus on structured spatial-spectral propagation. This decomposition makes measurement-consistent field estimation and physics-inspired field evolution complementary within a single-pass network.

Conclusion

We presented SMILE², a single-pass spectral-field estimation-evolution framework for CASSI reconstruction. SMILE² constructs measurement-aware CASSI fields, estimates a cleaner initial spectral field, and evolves it through a Fourier-domain damped-wave propagator. Experiments on simulation and real CASSI measurements demonstrate state-of-the-art end-to-end reconstruction quality with

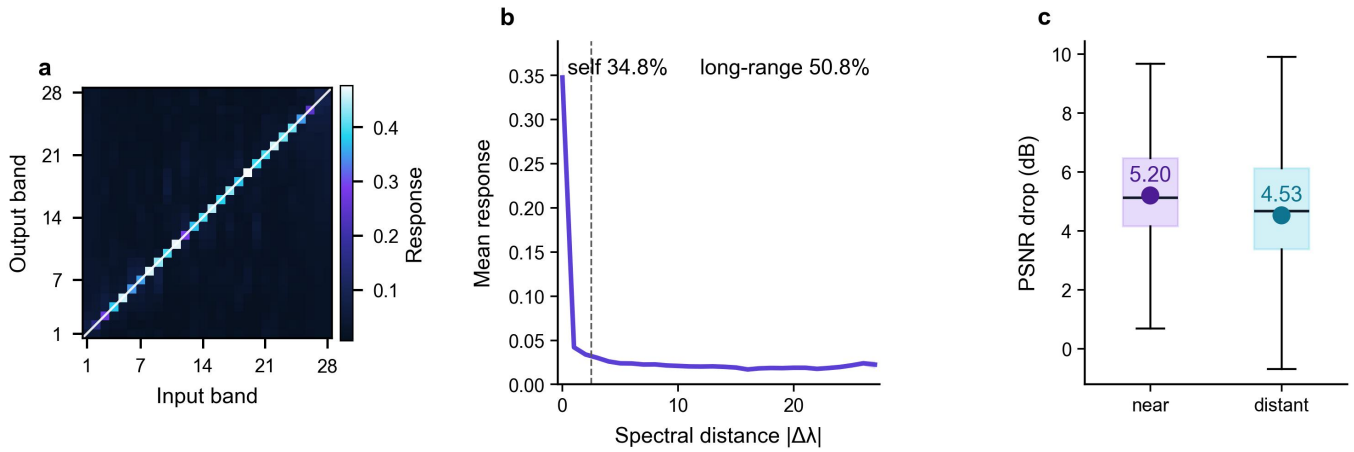


Figure 8: Mechanism analysis of cross-band propagation. The finite-difference mixing map and equal-count occlusion results reveal how SMILE² exploits spectral context.

Table 2: Ablation study under the SMILE²-S setting.

ID	Variant	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$
B0	SMILE ² -S	35.9378	0.9557	0.1116
A1	w/o SFESTIMATOR	35.4985	0.9482	0.1352
A2	w/o SFEVOLVER	24.3386	0.6160	0.4790
A3	w/o shifted-mask gating	35.3727	0.9473	0.1517
A4	w/o λ -derivative	35.5002	0.9484	0.1345

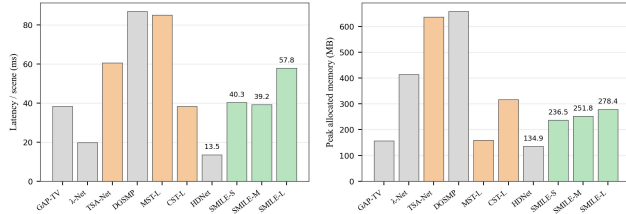


Figure 9: Latency and peak-memory comparison under the same test protocol.

compact computation. Mechanism analyses further show that SMILE² exploits structured cross-band propagation, providing a physically motivated alternative to generic token mixing for hyperspectral reconstruction.

Additional Diagnostics

Inference Latency and Peak Memory

Table 3 reports inference latency and peak allocated memory under the same test protocol. All methods are tested scene-by-scene at $256 \times 256 \times 28$ resolution. These measurements complement FLOPs and parameter counts, since actual throughput also depends on kernel implementation, memory access, and FFT scheduling.

Table 3: Inference latency and peak allocated memory on the TSA simulation benchmark.

Method	Params (M)	GFLOPs (G)	Time (ms)	Alloc. (MB)	Reserved (MB)
GAP-TV	4.27	75.63	38.23	156.17	194.00
λ -Net	32.73	28.40	19.71	414.00	500.00
TSA-Net	44.25	73.83	60.47	635.97	784.00
DGSM	3.76	646.36	86.89	658.29	774.00
MST-L	2.02	25.67	84.96	158.76	274.00
CST-L	3.00	20.34	38.25	316.45	388.00
HDNet	2.37	154.73	13.49	134.94	178.00
SMILE ² -S	1.02	28.28	40.26	236.47	274.00
SMILE ² -M	2.00	28.28	39.16	251.79	290.00
SMILE ² -L	3.23	38.60	57.82	278.42	316.00

Learned SWP Parameter Profiles

Figure 10 visualizes the learned SWP parameters in SMILE²-M. $\alpha(f_\lambda)$ and $v_s(f_\lambda)$ are computed from 28-mode SWP layers and thus correspond to spectral-mode-dependent damping and propagation speed. v_λ and t are module-level learnable scalars that control spectral second-derivative strength and evolution depth. Figure 11 further derives 3D damped-wave kernel visualizations from the learned average parameters under different evolution depths.

References

- Bioucas-Dias, J. M.; and Figueiredo, M. A. T. 2007. A New TwIST: Two-Step Iterative Shrinkage/Thresholding Algorithms for Image Restoration. *IEEE Transactions on Image Processing*, 16(12): 2992–3004.
- Bioucas-Dias, J. M.; Plaza, A.; Dobigeon, N.; Parente, M.; Du, Q.; Gader, P.; and Chanussot, J. 2012. Hyperspectral Unmixing Overview: Geometrical, Statistical, and Sparse Regression-Based Approaches. *IEEE Journal of Selected*

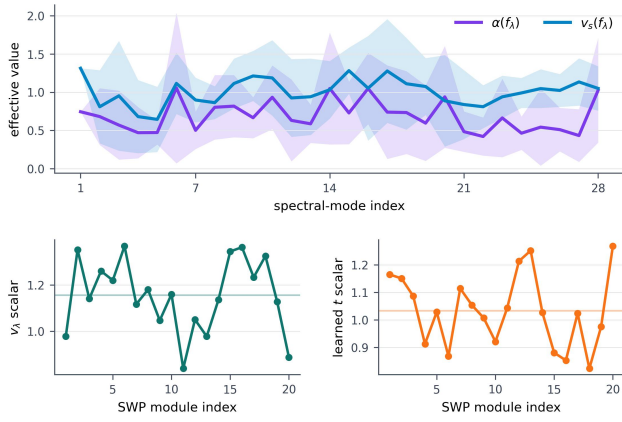


Figure 10: Learned spectral-mode parameters in SMILE²-M.

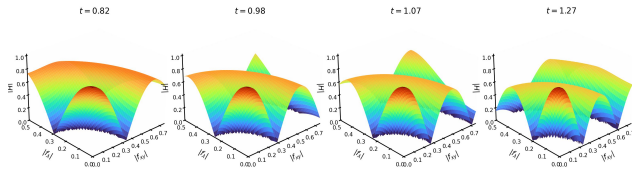


Figure 11: Derived 3D visualization of the learned damped-wave kernel under different evolution depths t .

Topics in Applied Earth Observations and Remote Sensing, 5(2): 354–379.

Cai, Y.; Lin, J.; Hu, X.; Wang, H.; Yuan, X.; Zhang, Y.; Timofte, R.; and Van Gool, L. 2022a. Coarse-to-Fine Sparse Transformer for Hyperspectral Image Reconstruction. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 686–704.

Cai, Y.; Lin, J.; Hu, X.; Wang, H.; Yuan, X.; Zhang, Y.; Timofte, R.; and Van Gool, L. 2022b. Mask-Guided Spectral-Wise Transformer for Efficient Hyperspectral Image Reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 17502–17511.

Cai, Y.; Lin, J.; Lin, Z.; Wang, H.; Zhang, Y.; Pfister, H.; Timofte, R.; and Van Gool, L. 2022c. MST++: Multi-Stage Spectral-Wise Transformer for Efficient Spectral Reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 744–754.

Cai, Y.; Lin, J.; Wang, H.; Yuan, X.; Ding, H.; Zhang, Y.; Timofte, R.; and Van Gool, L. 2022d. Degradation-Aware Unfolding Half-Shuffle Transformer for Spectral Compressive Imaging. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 37749–37761.

Chen, W.; Zha, Z.; Yuan, X.; Wang, S.; Wen, B.; Zhou, J.; Yan, G.; Fan, Z.; and Zhu, C. 2026. Phy-CoSF: Physics-Guided Continuous Spectral Fields Reconstruction and Super-Resolution for Snapshot Compressive Imaging. Accepted at ICML 2026, arXiv:2605.13583.

Cheng, Z.; Chen, B.; Lu, R.; Wang, Z.; Zhang, H.; Meng, Z.; and Yuan, X. 2022. Recurrent Neural Networks for Snapshot Compressive Imaging. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(2): 2264–2281.

Choi, I.; Jeon, D. S.; Nam, G.; Gutierrez, D.; and Kim, M. H. 2017. High-Quality Hyperspectral Reconstruction Using a Spectral Prior. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia 2017)*, 36(6): 218:1–218:13.

Cui, Y.; Liu, M.; Ren, W.; and Knoll, A. 2024. Hybrid Frequency Modulation Network for Image Restoration. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI)*, 722–730.

E, W. 2017. A Proposal on Machine Learning via Dynamical Systems. *Communications in Mathematics and Statistics*, 5(1): 1–11.

Gehm, M. E.; John, R.; Brady, D. J.; Willett, R. M.; and Schulz, T. J. 2007. Single-Shot Compressive Spectral Imaging with a Dual-Disperser Architecture. *Optics Express*, 15(21): 14013–14027.

Guibas, J.; Mardani, M.; Li, Z.; Tao, A.; Anandkumar, A.; and Catanzaro, B. 2022. Efficient Token Mixing for Transformers via Adaptive Fourier Neural Operators. In *International Conference on Learning Representations (ICLR)*. ArXiv:2111.13587.

Hu, X.; Cai, Y.; Lin, J.; Wang, H.; Yuan, X.; Zhang, Y.; Timofte, R.; and Van Gool, L. 2022. HDNet: High-Resolution Dual-Domain Learning for Spectral Compressive Imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 17542–17551.

Huang, T.; Dong, W.; Yuan, X.; Wu, J.; and Shi, G. 2021. Deep Gaussian Scale Mixture Prior for Spectral Compressive Imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 16216–16225.

Karniadakis, G. E.; Kevrekidis, I. G.; Lu, L.; Perdikaris, P.; Wang, S.; and Yang, L. 2021. Physics-Informed Machine Learning. *Nature Reviews Physics*, 3: 422–440.

Kong, L.; Dong, J.; Ge, J.; Li, M.; and Pan, J. 2023. Efficient Frequency Domain-Based Transformers for High-Quality Image Deblurring. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 5886–5895.

Kruse, F. A.; Lefkoff, A. B.; Boardman, J. W.; Heidebrecht, K. B.; Shapiro, A. T.; Barloon, P. J.; and Goetz, A. F. H. 1993. The Spectral Image Processing System (SIPS)—Interactive Visualization and Analysis of Imaging Spectrometer Data. *Remote Sensing of Environment*, 44(2–3): 145–163.

Lee-Thorp, J.; Ainslie, J.; Eckstein, I.; and Ontañón, S. 2022. FNet: Mixing Tokens with Fourier Transforms. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 4296–4313. Seattle, United States: Association for Computational Linguistics.

Li, J.; Zhao, W.; and Xu, T. 2026. Efficient Hyperspectral Image Reconstruction Using Lightweight Separate Spectral Transformers. arXiv:2601.01064.

- Li, Z.; Kovachki, N. B.; Azizzadenesheli, K.; Liu, B.; Bhat-tacharya, K.; Stuart, A. M.; and Anandkumar, A. 2021. Fourier Neural Operator for Parametric Partial Differential Equations. In *International Conference on Learning Representations (ICLR)*. ArXiv:2010.08895.
- Liu, Y.; Yuan, X.; Suo, J.; Brady, D. J.; and Dai, Q. 2019. Rank Minimization for Snapshot Compressive Imaging. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(12): 2990–3006.
- Luo, F.; Chen, X.; Guo, T.; Gong, X.; Zhang, L.; and Zhu, C. 2025. DSMT: Dual-Stage Multiscale Transformer for Hyperspectral Snapshot Compressive Imaging. *IEEE Transactions on Image Processing*, 34: 2473–2486.
- Meng, Z.; Jalali, S.; and Yuan, X. 2020. GAP-net for Snapshot Compressive Imaging. *arXiv preprint arXiv:2012.08364*.
- Meng, Z.; Ma, J.; and Yuan, X. 2020. End-to-End Low Cost Compressive Spectral Imaging with Spatial-Spectral Self-Attention. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 187–204.
- Miao, X.; Yuan, X.; Pu, Y.; and Athitsos, V. 2019. λ -Net: Reconstruct Hyperspectral Images from a Snapshot Measurement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 4059–4069.
- Rao, Y.; Zhao, W.; Zhu, Z.; Lu, J.; and Zhou, J. 2021. Global Filter Networks for Image Classification. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34. ArXiv:2107.00645.
- Shu, Z.; Wu, J.; Yan, W.; Liu, X.; Zhang, H.; Liu, C.; Mao, Y.; and Chen, J. 2026. WaveFormer: Frequency-Time Decoupled Vision Modeling with Wave Equation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 25428–25436.
- Simon, J.; Kunin, D.; Atanasov, A.; Boix-Adserà, E.; Bordelon, B.; Cohen, J.; Ghosh, N.; Guth, F.; Jacot, A.; Kamb, M.; Karkada, D.; Michaud, E. J.; Ottlik, B.; and Turnbull, J. 2026. There Will Be a Scientific Theory of Deep Learning. arXiv:2604.21691.
- Wagadarikar, A.; John, R.; Willett, R.; and Brady, D. 2008. Single Disperser Design for Coded Aperture Snapshot Spectral Imaging. *Applied Optics*, 47(10): B44–B51.
- Wang, J.; Li, K.; Zhang, Y.; Yuan, X.; and Tao, Z. 2025a. S2-Transformer for Mask-Aware Hyperspectral Image Reconstruction. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(6): 4299–4316.
- Wang, Z.; Liu, Y.; Tian, Y.; Liu, Y.; Wang, Y.; and Ye, Q. 2025b. Building Vision Models upon Heat Conduction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9707–9717.
- Willard, J.; Jia, X.; Xu, S.; Steinbach, M.; and Kumar, V. 2023. Integrating Scientific Knowledge with Machine Learning for Engineering and Environmental Systems. *ACM Computing Surveys*, 55(4): 1–37.
- Yao, Z.; Liu, S.; Yuan, X.; and Fang, L. 2024. SPECAT: SPatial-spEctral Cumulative-Attention Transformer for High-Resolution Hyperspectral Image Reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 25368–25377.
- Yasuma, F.; Mitsunaga, T.; Iso, D.; and Nayar, S. K. 2010. Generalized Assorted Pixel Camera: Postcapture Control of Resolution, Dynamic Range, and Spectrum. *IEEE Transactions on Image Processing*, 19(9): 2241–2253.
- Yuan, X. 2016. Generalized Alternating Projection Based Total Variation Minimization for Compressive Sensing. In *2016 IEEE International Conference on Image Processing (ICIP)*, 2539–2543.
- Zhang, J.; Zeng, H.; Cao, J.; Chen, Y.; Yu, D.; and Zhao, Y.-P. 2024a. Dual Prior Unfolding for Snapshot Compressive Imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 25742–25752.
- Zhang, J.; Zeng, H.; Chen, Y.; Yu, D.; and Zhao, Y.-P. 2024b. Improving Spectral Snapshot Reconstruction with Spectral-Spatial Rectification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 25817–25826.